Practical Numerical Simulations HW2

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1 Shooting Method

We consider the second order ODE with boundary conditions,

$$\frac{d^2x}{dt^2} = -\frac{tx(t+2)}{2+t^2x^2}, \quad x(0) = \frac{3}{4}, \quad x(10) = -1$$

To solve this ODE we use the 4th order Runge-Kutta method together with the shooting method, which helps us find the correct initial slope $x'(0) = s$ that satisfies the boundary conditions.

The Runge-Kutta method was employed as follows:

$$v = \frac{dx}{dt} \Rightarrow a(t, x, v) = \frac{dv}{dt} = -\frac{tx(t+2)}{2+t^2x^2}$$

We run the Runge-Kutta method for $N = 5000$ iterations with a step size of $h = \frac{t_f - t_0}{N} = 0.002$ using initial values $t_0 = 0, x_0 = \frac{3}{4}, v_0 = x'(0) = s$. The first iteration stage values are then defined as:

$$k_1 = hv_0 = hs$$

$$l_1 = h\,a(t_0, x_0, v_0)$$

$$k_2 = hv_1 = h\left(v_0 + \frac{l_1}{2}\right)$$

$$l_2 = h\,a(t_1, x_1, v_1) = h\,a\left(t_0 + \frac{h}{2}, x_0 + \frac{k_1}{2}, v_0 + \frac{l_1}{2}\right)$$

$$k_3 = hv_2 = h\left(v_1 + \frac{l_2}{2}\right)$$

$$l_3 = h\,a(t_2, x_2, v_2) = h\,a\left(t_0 + \frac{h}{2}, x_0 + \frac{k_2}{2}, v_0 + \frac{l_2}{2}\right)$$

$$k_4 = hv_3 = h\,(v_2 + l_3)$$

$$l_4 = h\,a(t_3, x_3, v_3) = h\,a(t_0 + h, x_2 + k_3, v_2 + l_3)$$

The initial values are then updated as follows:

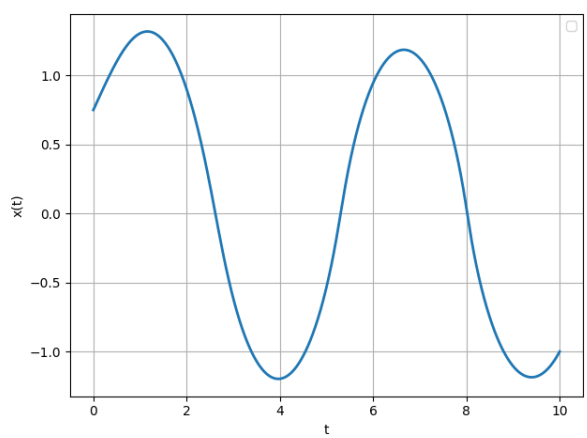
$$t_1 = t_0 + h$$
$$x_1 = x_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$
$$v_1 = v_0 + \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4)$$

All of the above steps are then repeated again now starting with x_1, v_1, t_1 as our initial values. This is done 5000 times until we reach $t_f = 10$.

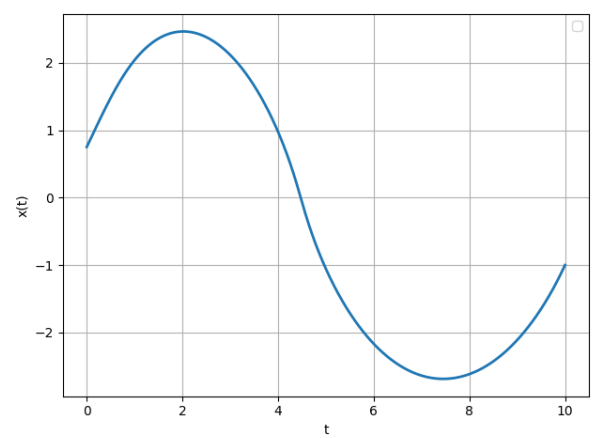
The shooting method is implemented as,

$$F(s) = x(10; s) - x(10) = x(10; s) + 1$$

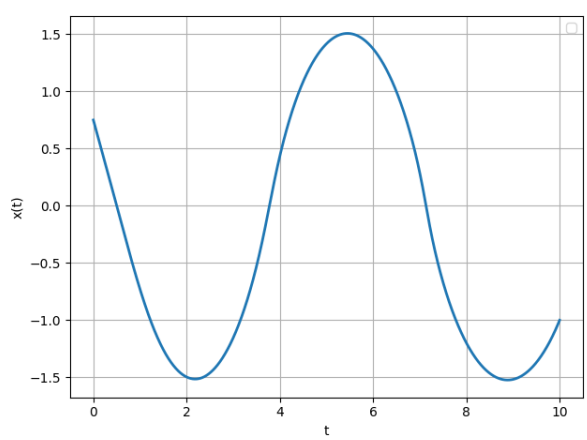
where $x(10; s)$ is the value our Runge-Kutta method outputs given a specific input of s. We want $F(s) = 0$. We define an initial starting point from where we want to start finding suitable values of s at $x = -5.0$ and step along the x-axis in steps of 0.2. The bisection method was applied in each of these intervals for 100 iterations to locate any roots.



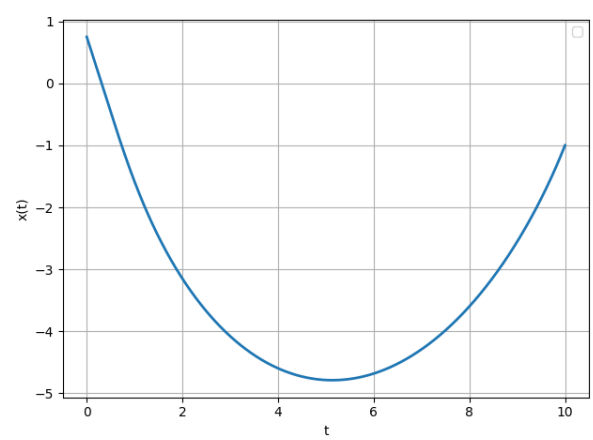
(a) $s = 0.742301$



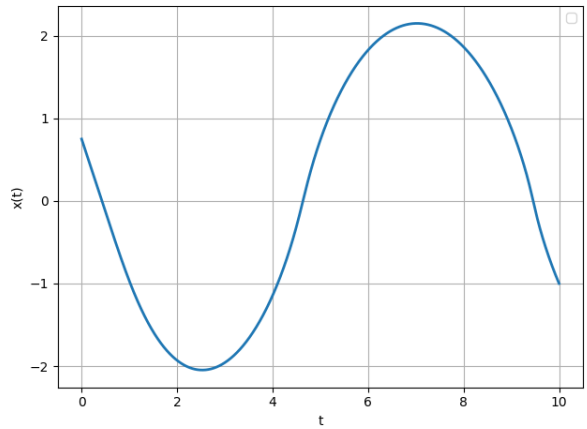
(b) $s = 1.50569$



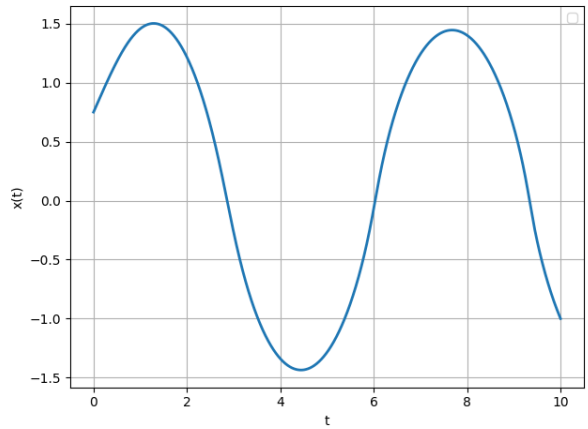
(a) $s = -1.48228$



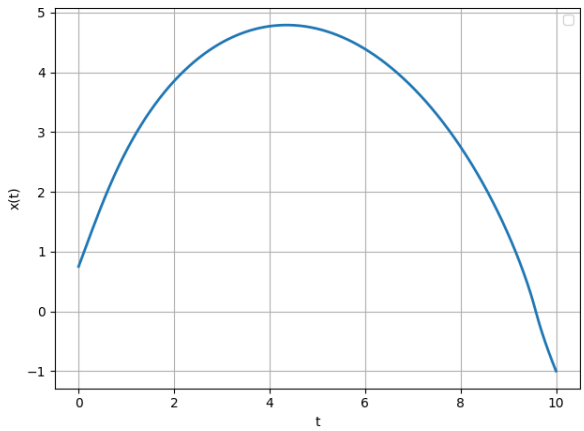
(b) $s = -2.41626$



(a) $s = -1.74799$



(b) $s = 0.902091$



(a) $s = 2.17010$

Figure 4: Solutions for $x(t)$ for different values of s using the shooting method.

The shooting method can give multiple values of s. The differential equation for $x(t)$ defines a family of possible trajectories, while the boundary conditions $x(0) = \frac{3}{4}$ and $x(10) = -1$ select the specific trajectories which are mathematically valid. The values of $x'(0) = s$ which satisfy the boundary conditions were found correct to 6 s.f. to be $s = 0.742301$, 1.50569, -1.48228, -2.41626, -1.74799, 0.902091, 2.17010.

2 Newtonian Gravity

The leap frog method was used to compute the trajectories and final positions of the system of four planets at $t = 5$.

Plant	**Mass**	**Initial Position** (x_0, y_0)	**Initial Velocity** $(v_{x,0}, v_{y,0})$
1	2.2	$-0.50, 0.10$	$-0.84, 0.65$
2	0.8	$-0.60, -0.20$	$1.86, 0.70$
3	0.9	$0.50, 0.10$	$-0.44, -1.50$
4	0.4	$0.50, 0.40$	$1.15, -1.60$

Table 1: Initial positions and velocities of planets

The planets move according to Newton's inverse square law of gravity, which states that the force of attraction exerted on planet i due to planet j has magnitude,

$$|F_{ij}| = \frac{Gm_i m_i}{r_{ij}^2}, \quad G = 1$$

Newton's third law of motion tells us that for every action, there is an equal and opposite reaction. This principle is useful for our simulation as it means the gravitational force on planet i due to j is equal and opposite in magnitude to the force exerted on planet j due to i, i.e.

$$\vec{F_{ij}} = -\vec{F_{ji}}$$

This means that both accelerations can be computed within the same loop iteration in our program. The acceleration on planet i is computed as follows:

$$m_i \vec{a_i} = \sum_{j \neq i} \vec{F_{ij}} = \sum_{j \neq i} \frac{m_i m_j (\vec{r_j} - \vec{r_i})}{|\vec{r_j} - \vec{r_i}|^3}$$

$$\vec{a_i} = \sum_{j \neq i} \frac{m_i (\vec{r_j} - \vec{r_i})}{|\vec{r_j} - \vec{r_i}|^3}$$

Splitting this into x and y components we get:

$$a_{x,i} = \sum_{j \neq i} \frac{m_i (x_j - x_i)}{|\vec{r_j} - \vec{r_i}|^3}$$

$$a_{y,i} = \sum_{j \neq i} \frac{m_i (y_j - y_i)}{|\vec{r_j} - \vec{r_i}|^3}$$

We can now use the leap frog integration method to compute the trajectories of each planet and find each of their positions at $t = 5$. For planet i at a step n the leap frog method is defined as:

$$\vec{v_i}^{n+\frac{1}{2}} = \vec{v_i} + \frac{h}{2} \vec{a_i}^n$$

$$\vec{r_i}^{n+1} = \vec{r_i}^n + h \vec{v_i}^{n+\frac{1}{2}}$$

$$\vec{a_i}^{n+1} = \vec{a_i}(\vec{r_i}^{n+1})$$

$$\vec{v}_i^{n+1} = \vec{v}_i^{n+\frac{1}{2}} + \frac{h}{2}\vec{a}_i^{n+1}$$

The step size is defined as $h = \frac{t_f - t_0}{N}$ where we are computing $N = 50000$ iterations.

The evolution of the system of planets from $t_0 = 0$ to $t_f = 5$ was found to be:

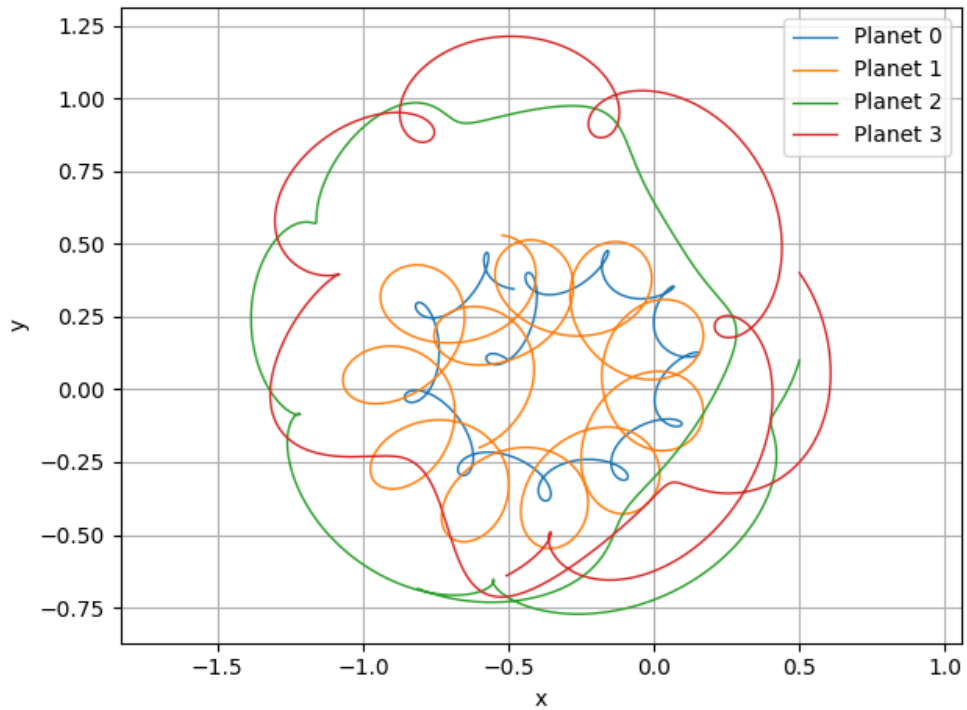


Figure 5: Trajectories of each planet when $t = 5$

Planet	**Position** (x, y)
1	$-0.48159, 0.34500$
2	$-0.52176, 0.52920$
3	$-0.81146, -0.68449$
4	$-0.50697, -0.64079$

Table 2: Final positions of planets at $t = 5$ for $N = 50000$

Planet	**Position** (x, y)
1	$-0.48158, 0.34500$
2	$-0.52177, 0.52920$
3	$-0.81146, -0.68449$
4	$-0.50697, -0.64080$

Table 3: Final positions of planets at $t = 5$ for $N = 100000$

To verify the final positions of each planet are accurate to 4 s.f. the leap frog method was simulated

for $N = 50000$ and $N = 100000$ iterations. In order for the location of the planets at $t = 5$ to be accurate to 4 s.f. the error between positions after N_1 and N_2 iterations should be less than 5×10^{-5}, i.e. for the x component:

$$\left|x_{N_1} - x_{N_2}\right| < 5 \times 10^{-5}$$

The final positions of all four planets were found to agree to within 4 s.f. for both simulations of the system. This indicates the positions at $t = 5$ have converged with respect to the step size and are accurate to at least 4 s.f.